

120 Days of Quant Finance

Month 2: Derivatives & Time Series

Day 33: Hedging Using Option Greeks

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Introduction

Option positions are exposed to multiple sources of risk, including changes in the underlying asset price, time decay, volatility, and interest rates. Hedging using Greeks provides a structured framework to identify and manage these risks.

This section introduces the principles of Greek-based hedging and explains how Delta, Gamma, Theta, Vega, and Rho are used in practice.

Taylor Approximation of Option Value

For small changes in market variables, the change in option value can be approximated using a Taylor expansion:

$$\Delta V \approx \Delta \Delta S + \frac{1}{2} \Gamma (\Delta S)^2 + \Theta \Delta t + \nu \Delta \sigma + \rho \Delta r.$$

This decomposition forms the basis of Greek-based risk management.

Delta Hedging

Delta hedging aims to eliminate first-order exposure to changes in the underlying asset price. A portfolio with Delta equal to zero is locally insensitive to small price movements.

Delta hedging is typically achieved by taking an offsetting position in the underlying asset.

Gamma Risk and Dynamic Hedging

Gamma measures the sensitivity of Delta to changes in the underlying price. When Gamma is non-zero, Delta changes as the market moves, requiring frequent rebalancing of the hedge.

This process is known as dynamic hedging and introduces transaction costs and residual risk.

Theta and the Cost of Carry

Theta captures the effect of time decay on option value. Long option positions typically experience negative Theta, while short option positions benefit from time decay.

There is an inherent tradeoff between Gamma and Theta, as positions with high Gamma generally suffer from greater time decay.

Vega Hedging

Vega measures sensitivity to changes in volatility. Volatility risk is often a dominant driver of option portfolio performance.

Vega hedging involves combining options with offsetting volatility exposure to reduce sensitivity to volatility shocks.

Rho Hedging

Rho measures sensitivity to interest rate changes. While often small for short-dated equity options, Rho becomes significant for long-dated and interest-rate-sensitive derivatives.

Practical Hedging Considerations

In practice, it is not possible to hedge all risks simultaneously without incurring significant costs. Risk managers prioritize hedging based on market conditions, liquidity, and strategic objectives.

Greek-based hedging is therefore a process of risk control rather than complete risk elimination.

Conclusion

Option Greeks provide a powerful framework for decomposing and managing risk in derivative portfolios. Effective hedging requires understanding the dynamic nature of Greeks and the tradeoffs between different sources of risk.